

Module 01 Data Integration Fundamentals Workshop

Ronny Zakaria

Technical Leader - Analytic Platform

IBM Analytics – Indonesia

ronnyz@id.ibm.com

IBM InfoSphere Information Server for Data Integration



Important links

- IBM Education offerings
 - Additional education available for customer and partners
 - <http://www-304.ibm.com/jct03001c/services/learning/ites.wss/us/en?pageType=page&c=a0000352>

- IBM DeveloperWorks
 - Technical articles and discussions
 - <http://www.ibm.com/developerworks/data/>

- IBM Redbooks
 - Deep knowledge and guidance for IBM products
 - <http://www.redbooks.ibm.com/>

Module 01 InfoSphere Portfolio & Data Integration Components

IBM InfoSphere Information Server for Data Integration

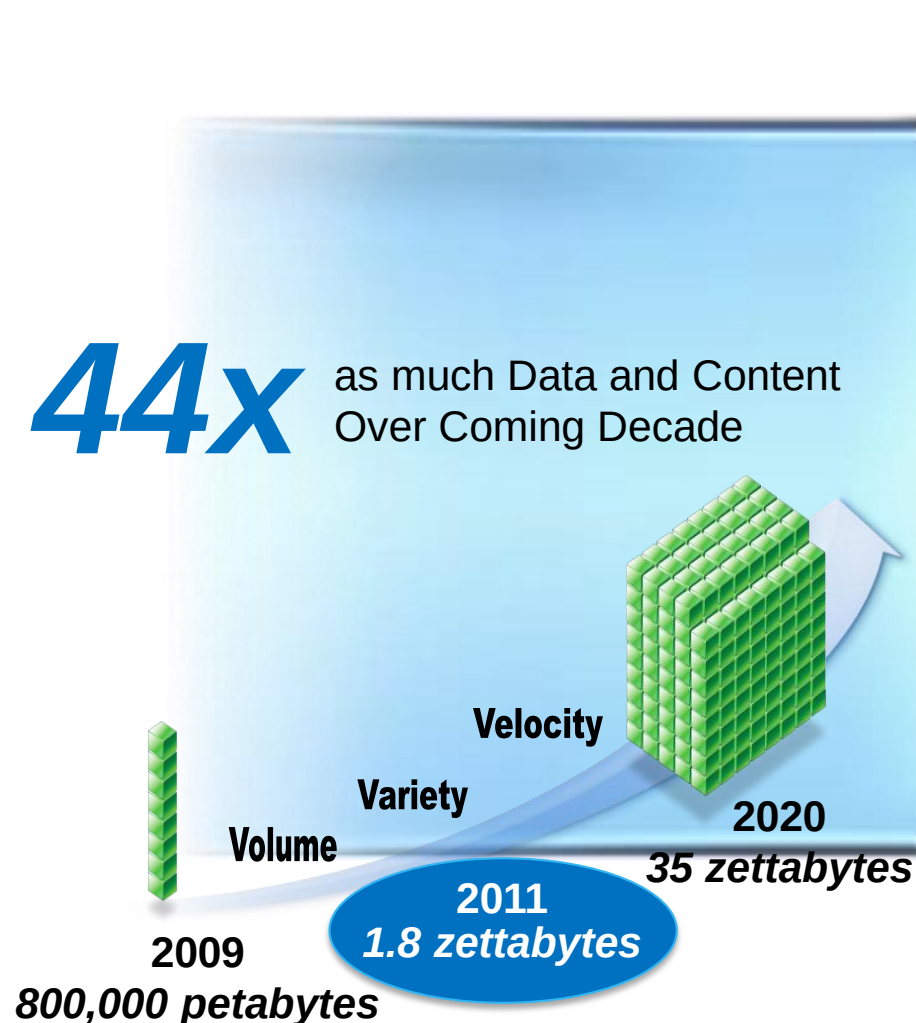


Module Objectives

After completing this topic, you should be able to:

- Describe the key components of InfoSphere Information Server
- Understand how to position each InfoSphere Information Server component
- Understand InfoSphere Information Server packages and the tools associated with each package

Data is growing ... So is the need to take control



- Data integration can help you take charge of diverse and changing data

Value	Metric
1000	kB kilobyte
1000 ²	MB megabyte
1000 ³	GB gigabyte
1000 ⁴	TB terabyte
1000 ⁵	PB petabyte
1000 ⁶	EB exabyte
1000 ⁷	ZB zettabyte
1000 ⁸	YB yottabyte

Integrating information from across the enterprise

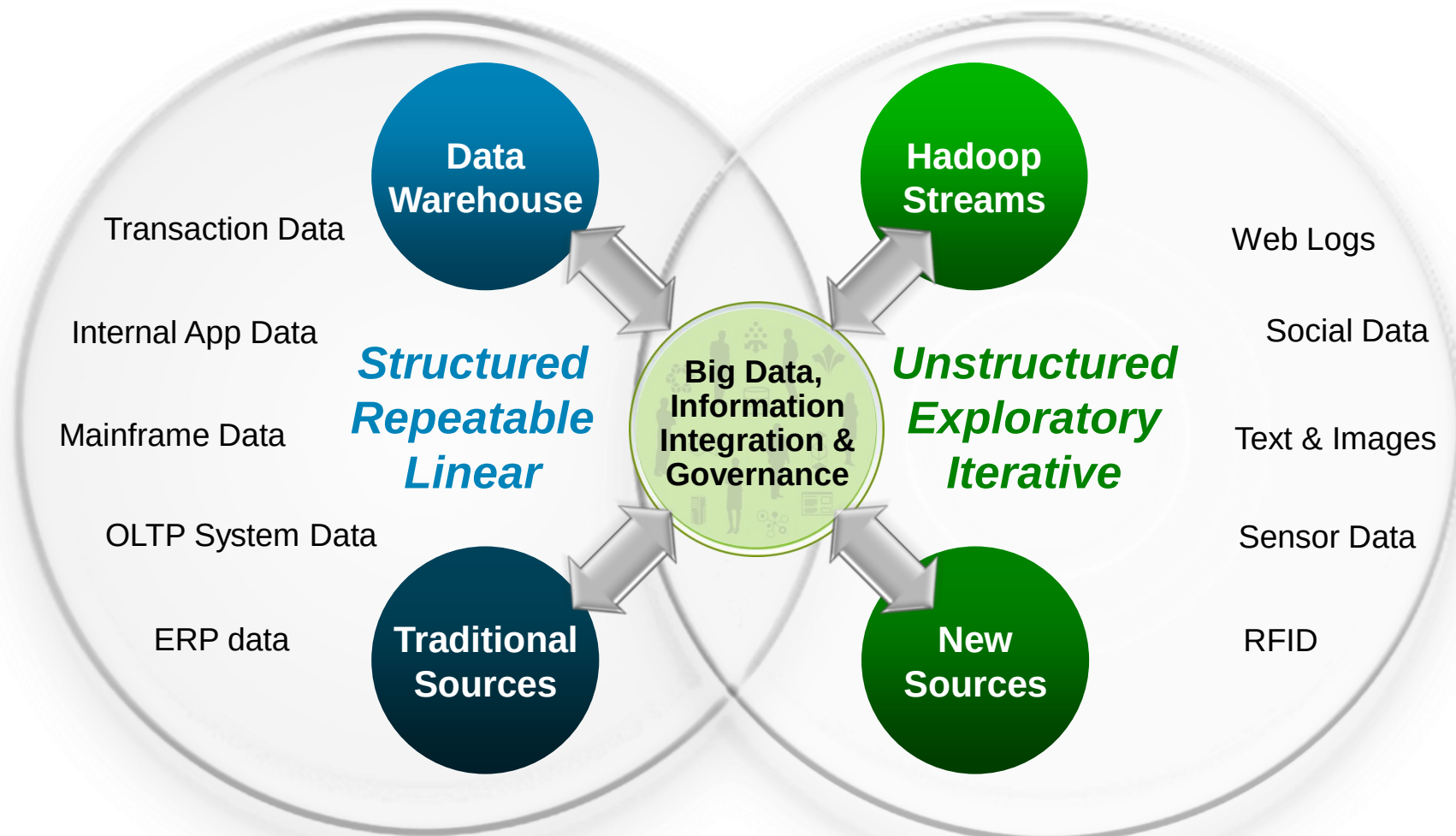
Traditional Approach

Structured, analytical, logical



New Approach

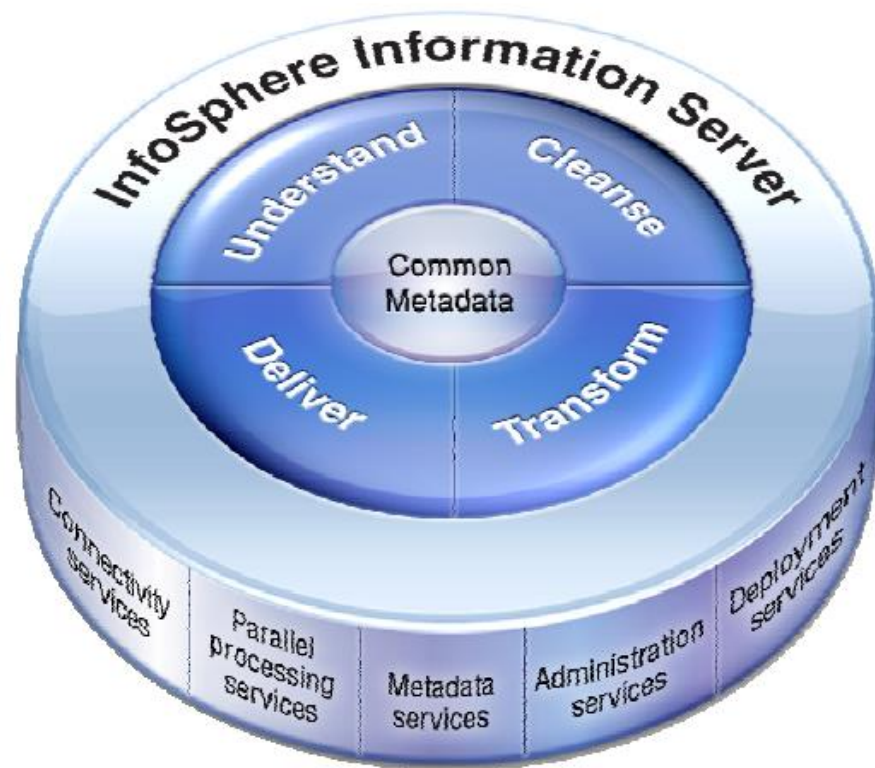
Creative, holistic, intuitive



IBM InfoSphere Information Server

Everything you need to integrate heterogeneous information and deliver it when and where it is needed

- **Market-leading data integration platform**
- Linear scalability for exploding data volumes
- Complete metadata integration and data lineage insight due to common repository
- Aligns business and IT objectives
- Optimized connectivity to diverse data sources



InfoSphere Information Server Package

Integrating and transforming data and content to deliver accurate, consistent, timely and complete information on a single platform unified by a common metadata layer



Business Information Exchange

Understanding & Collaboration

- Information blueprints
- Relationship discovery across data sources
- IT-to-business mapping



Data Quality

Cleansing & Monitoring

- Analysis & validation
- Data cleansing
- Data quality rules & management



Data Integration

Transformation

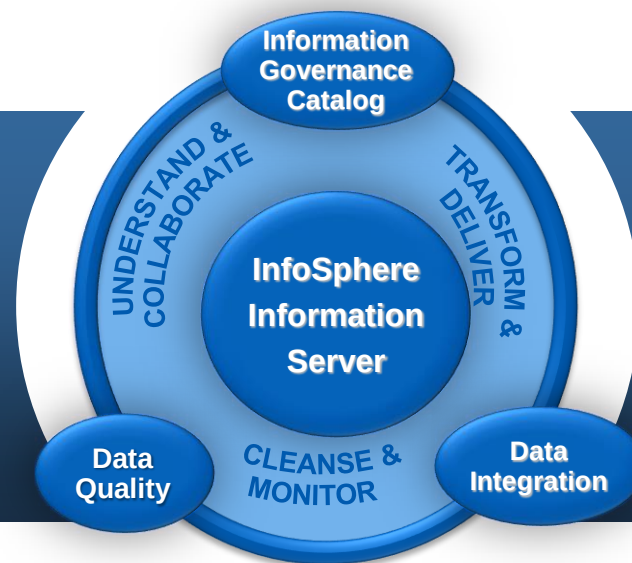
- Massive scalability
- Power for any complexity
- Total traceability

Delivery

- Data capture at any time
- Delivery anywhere
- Big data readiness

New capabilities in Information Server v11.5

Integrating and transforming data and content to deliver accurate, consistent, timely and complete information on a single platform unified by a common metadata layer



Information Governance Catalog

Understand & Collaborate

- Catalog technical metadata & align w/ business language
- Manage (big) data lineage
- **New** compliance reporting

NEW

Data Quality

Cleanse & Monitor

- Analyze & validate w/ **enhanced** classification
- Cleanse & standardize
- Define, manage & monitor data rules + exceptions

NEW

Data Integration

Transform & Deliver

- Massive scalability
- Power for any complexity
- Deliver in batch and/or real-time with change capture

NEW

- common connectivity • shared metadata • security (**new** data privacy functions included)
- common execution engine with flexible deployments (**new** native on YARN)

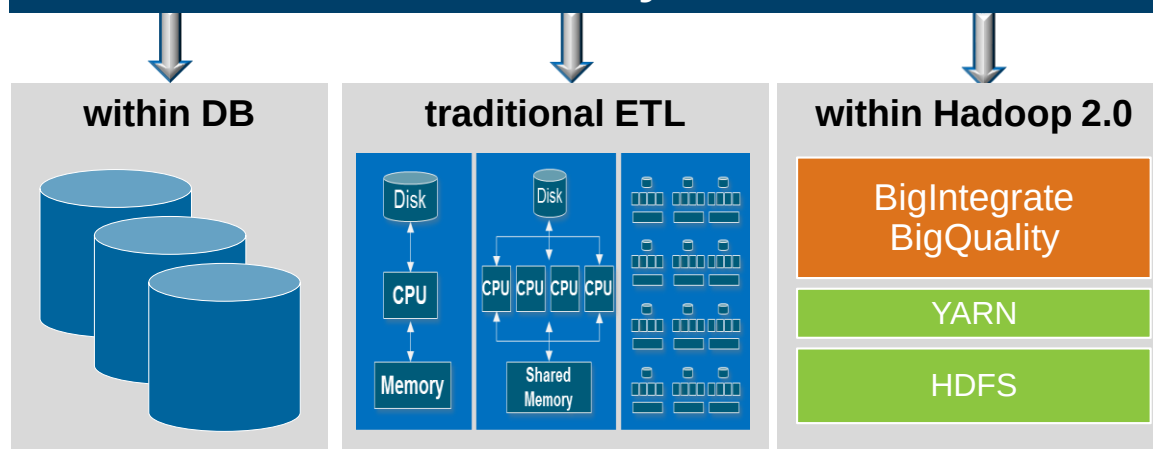
IBM's Unique Approach: Information Integration & Governance for the Enterprise



- Highest performance through native execution of Information Server engine directly on Yarn
- Design your transformation or cleansing once and run it on your Hadoop Cluster, your traditional engine or within your database

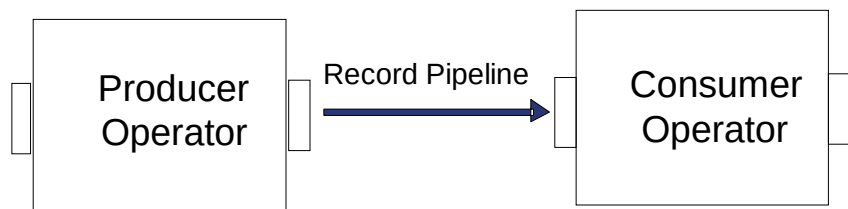
One Integration & Quality Design

Execute "Anywhere"



Information Server is upwards of 15x faster than M/R

InfoSphere Information Server is designed to exploit data pipelining to minimize I/O and maximize overall performance.

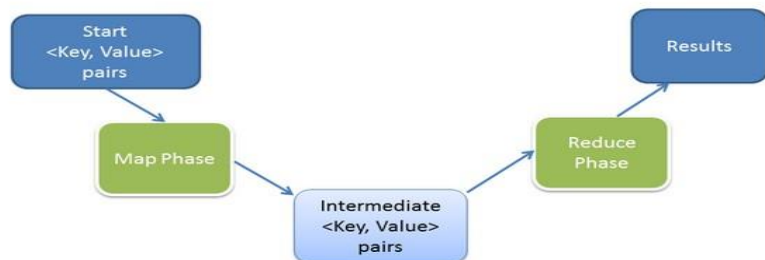


- Data is streamed from producer to consumer with data repartitioning without landing to disk
 - Intermediate data is not written to disk
- Flow control prevents the producer from overrunning the consumer
- Pipelined and partitioned parallelism provides efficiency and high performance

Job-level Fault Tolerance

Map-Reduce is designed for task-level fault tolerance: The downside of this is significantly lower performance.

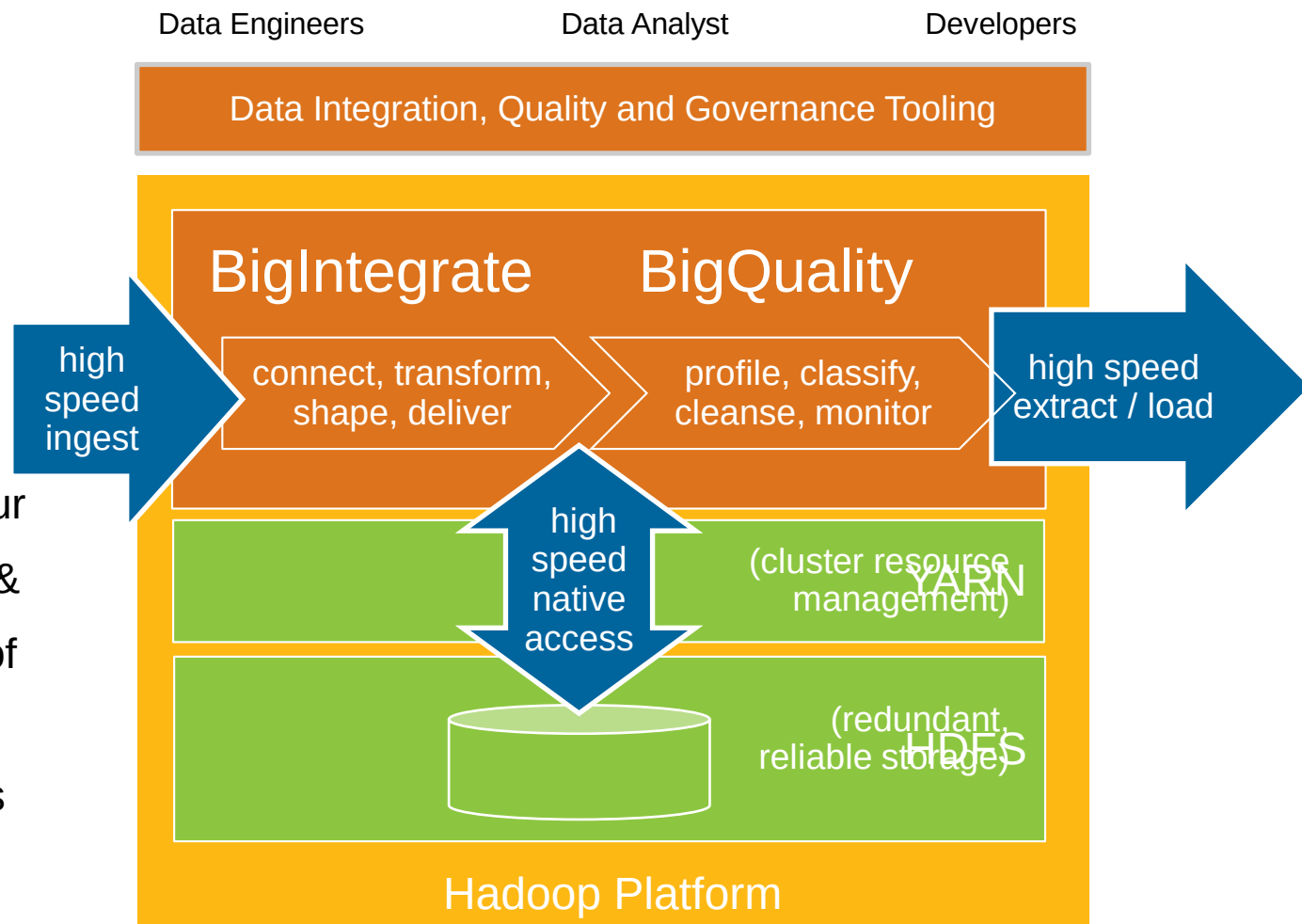
Task-level Fault Tolerance



- Producer (Map) writes intermediate results to disk
 - Consumer (Reduce) “pulls” the data
- This design provides task-level fault tolerance
 - This design lowers overall performance and efficiency by orders of magnitude

IBM BigInsights BigIntegrate & BigQuality

- The most scalable Transformation and Data Integration and Quality engine now runs natively on Hadoop
- Get enterprise-class transformation and cleansing for your Hadoop data
- Use the power of your Hadoop cluster to integrate, transform & cleanse data without writing a single line of code
- Hadoop distribution currency: BigInsights 4.0, Open Data Platform, HortonWorks 2.2, Cloudera 5.3



Data Integration on Hadoop

Ingest, transform, process and deliver any data into & within Hadoop

Satisfy the most complex transformation requirements with the most scalable runtime available in batch or real-time

Connect

- **Scalable transformation on the data nodes of the hadoop cluster**
- **File Connector improvements** (column level lineage, fixed length file support)
- **SQL Srv 2014, Sybase ASE/IQ 16, Teradata 15.10**



Design & Transform

- Transform and aggregate any data volume
- Benefit from hundreds of built-in transformation functions
- Leverage metadata-driven productivity and enable collaboration
- **Embedded protection of sensitive data in flight leveraging best-in class Optim Masking** (Masking included in Information Server Enterprise Ed.)



Manage & Monitor

- Use a simple, web-based dashboard to manage your runtime environment



More capabilities included in the base License

- We will discontinue to sell the following products as add-on offerings starting with Information Server v11.5*
 - DataStage Balanced Optimization*
 - Information Server Packs for:
 - Salesforce
 - Oracle Apps
 - PeopleSoft
 - Siebel
 - JD Edwards
 - Hyperion
- Those capabilities will be rolled into the core license of DataStage and Information Server Editions

**Note: we have opted not to withdraw these for now to ensure customers can still license these when on older Information Server releases*

Business Information Exchange: *Understand your information, transforming it into an enterprise asset*

Three dimensions of understanding and governance:

Business

Gain & manage business perspective about information and align with IT

- ★ *leading technology for business-friendly access & pre-packaged terms*
- ★ *time-saving Industry Models for warehouses in key industries*

Process

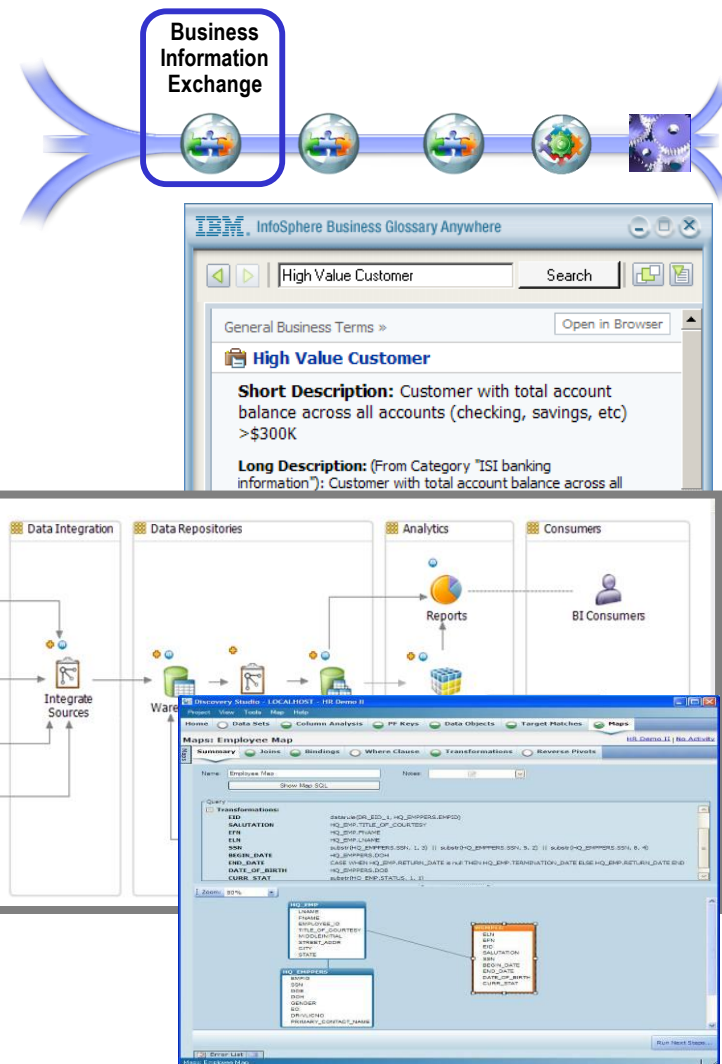
Guide projects with best practices to achieve goals with reduced risk

- ★ *unique capability to architect information projects with embedded methodology that can be tracked*

Technology

Discover data structures and understand your lineage to manage compliance

- ★ *unique capability for discovering business objects*



Data Quality: Create & maintain quality data, turning data assets into trusted information

Analyze data, cleanse data and control data quality

Analyze

Use source system analysis to understand your issues

- ★ *automated discovery of critical data and hidden data relationships*

Cleanse

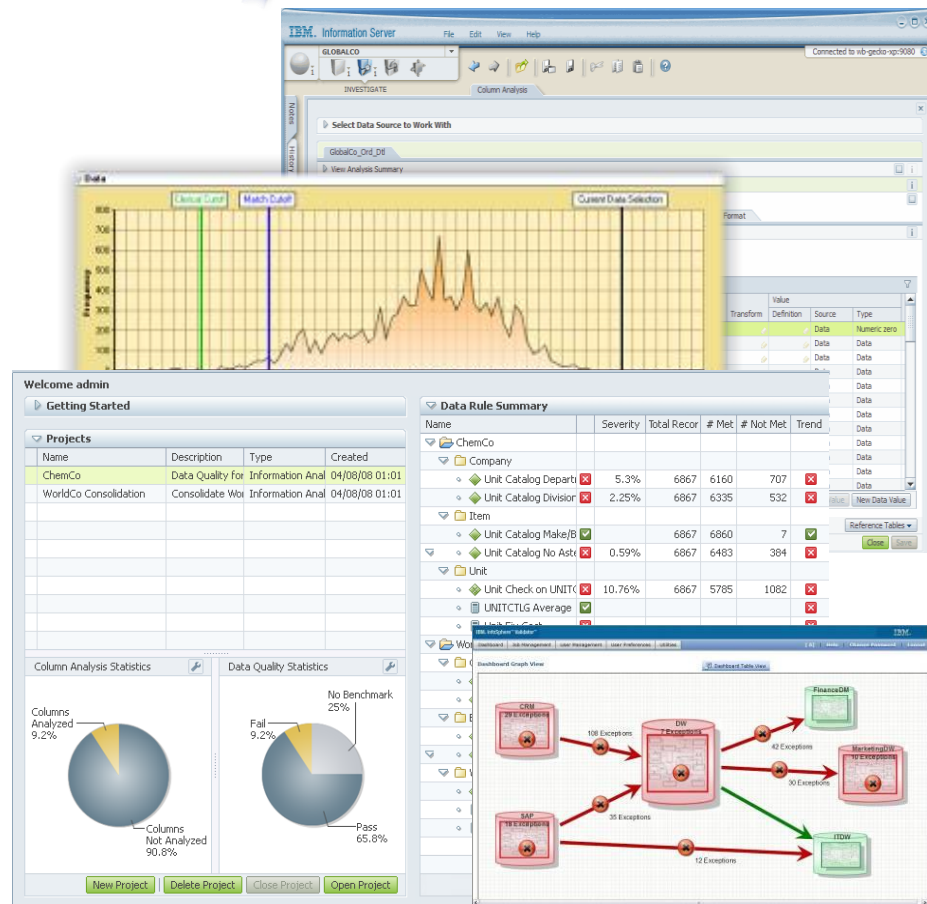
Investigate, standardize, match and survive data

- ★ *most comprehensive & customizable solution*

Control & monitor quality

Assess and monitor the quality of your data in any place (database or data flow) and across systems

- ★ *unique capability to align DQ metrics with business & governance objectives*

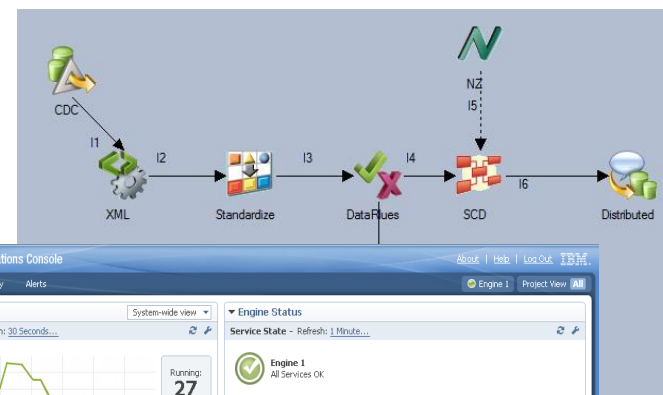
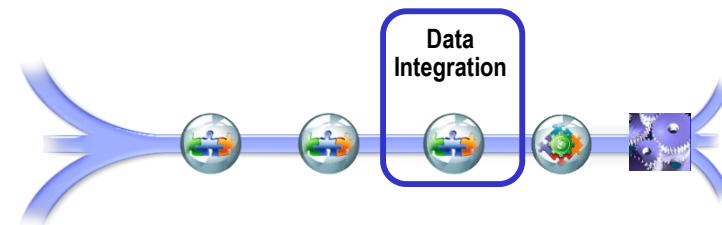


Integration: Transform and deliver to any system, improving time to value

Integrate and transform data on demand across multiple sources and targets ...

Transform

- ★ Satisfy the most complex transformation requirements with the most scalable runtime available
- Transform and aggregate any data volume
- Benefit from hundreds of built-in transformation functions
- Leverage metadata-driven productivity and enable collaboration
- Use a simple, web-based dashboard to manage your runtime environment
- Manage your requirements for transformation activities to align with the business



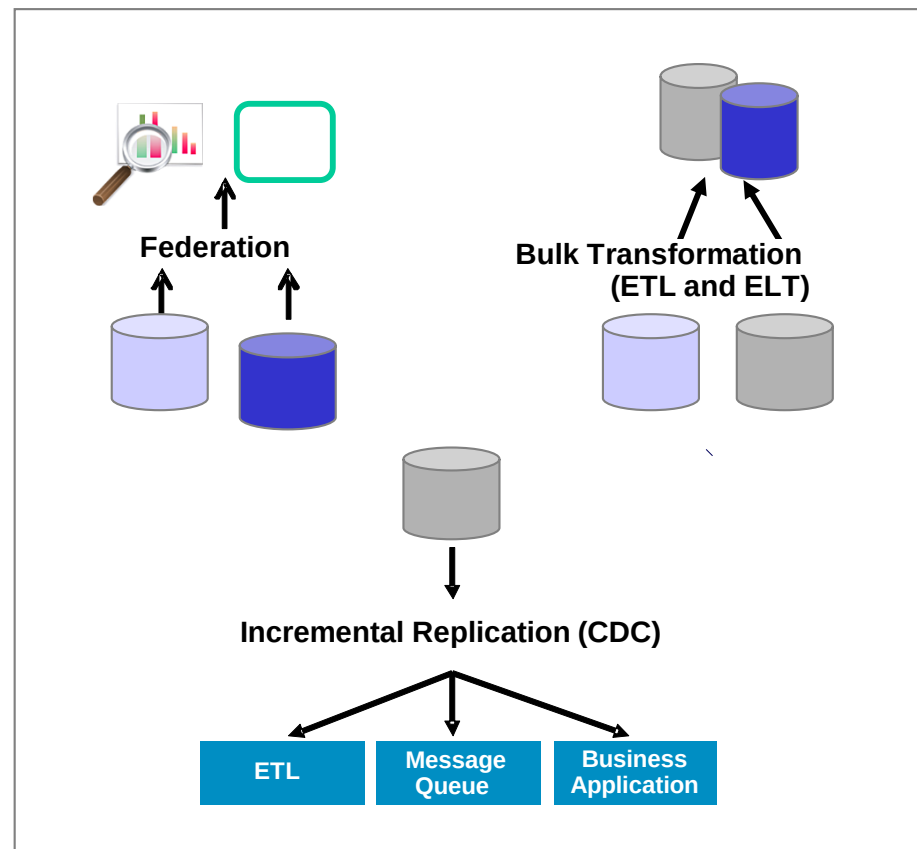
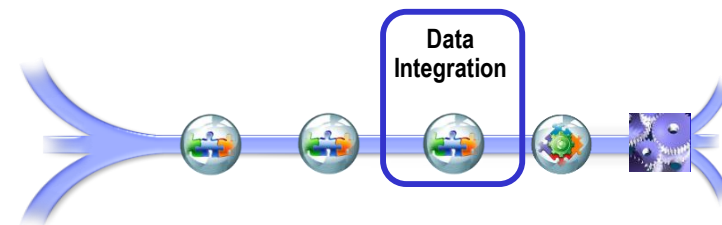
Integration: *Transform and deliver to any system, improving time to value*

Deliver data efficiently according to your business requirements

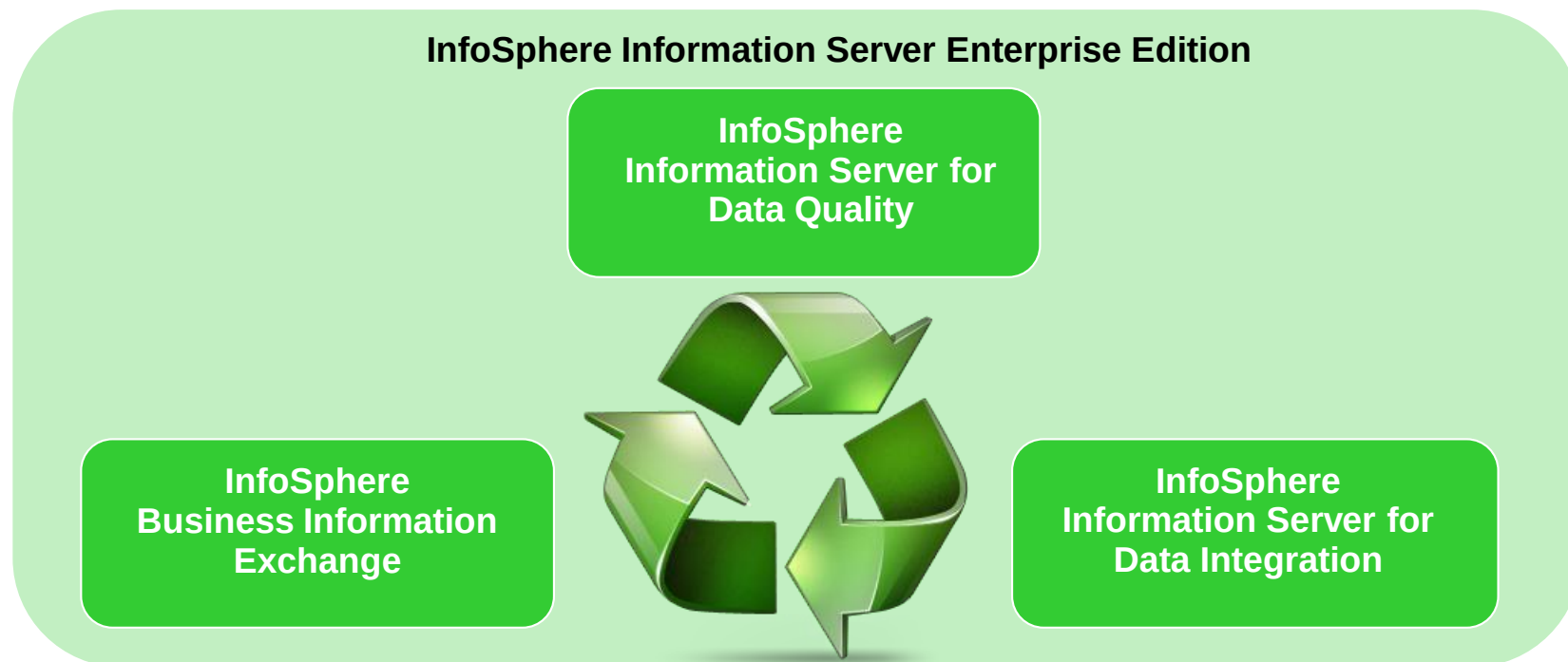
Deliver

★ *Leverage unique capabilities to:*

- Read from low-impact DBMS logs
- Apply any level of complex data integration/data quality logic
- Guarantee delivery through two-phased commit to one or more DBMS or MQ targets
- Support the most complex & challenging real-time integration requirements



InfoSphere Information Server Packages

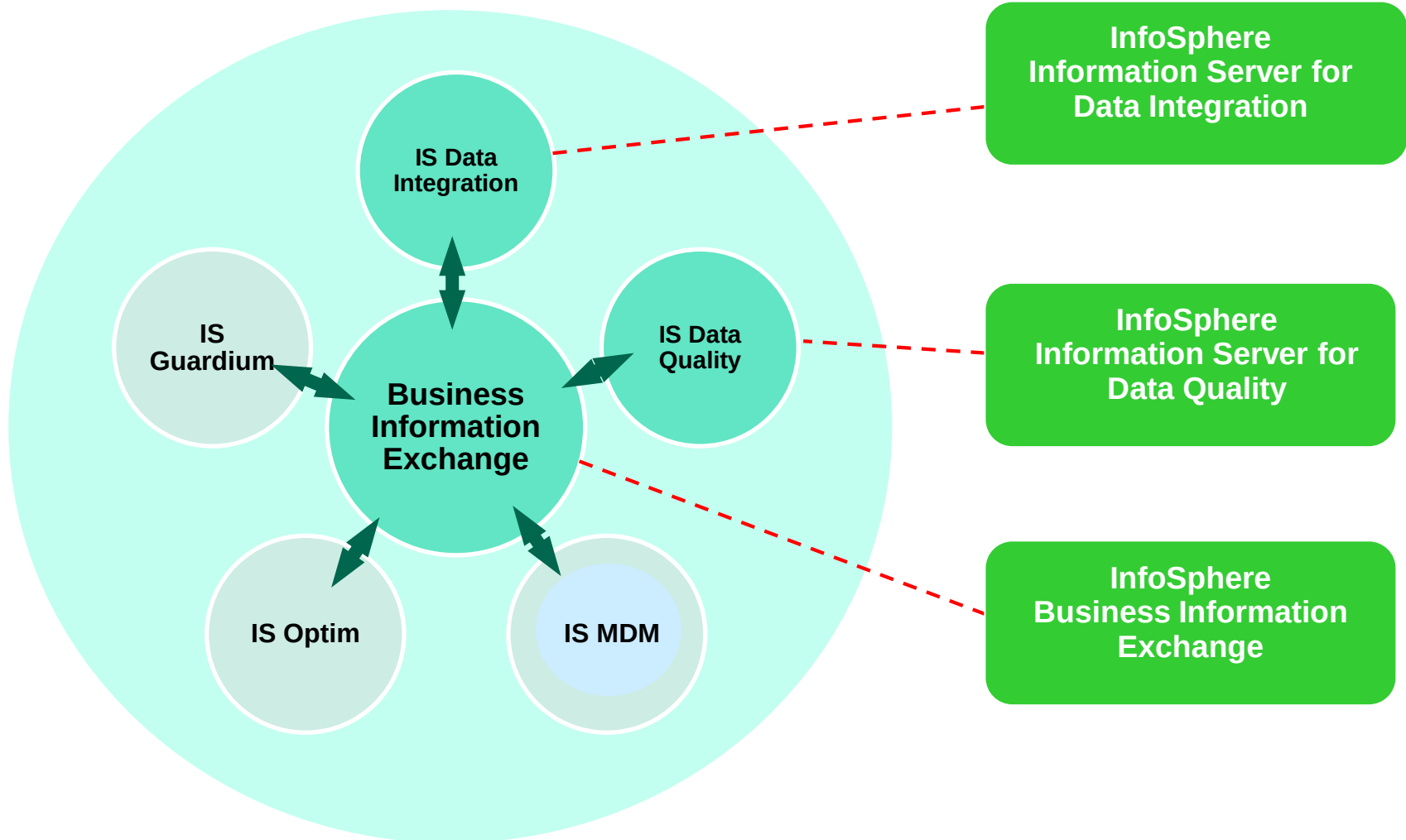


InfoSphere Information Server made up of 4 packages:

- ✓ InfoSphere Business Information Exchange
- ✓ InfoSphere Information Server for Data Integration
- ✓ InfoSphere Information Server for Data Quality
- ✓ InfoSphere Information Server Enterprise Edition (*inclusive of all 3 packages above*)

Information Server Strategic Fit

IBM InfoSphere Platform
 “Information Integration and Governance”



Detailed Capability Comparison

	InfoSphere Information Governance Catalog	InfoSphere Information Server For Data Integration	InfoSphere Information Server For Data Quality	InfoSphere Information Server Enterprise Edition	BigInsights BigIntegrate	BigInsights BigQuality
Business Glossary	✓	✓ ¹	✓ ¹	✓	✓ ¹	✓ ¹
Metadata Management and Lineage	✓	✓ ¹	✓ ¹	✓	✓ ¹	✓ ¹
Logical and Physical Data Modeling	✓	✓		✓	✓	
Data Cleansing and Enrichment			✓	✓		✓
Data Quality Validation & Monitoring			✓	✓		✓
Data Stewardship	✓			✓		
SOA Deployment		✓	✓	✓	✓	✓
Data Specification Mapping		✓		✓	✓	
Extract, transform, load (ETL)		✓		✓	✓	
Change Data Delivery		✓ ²		✓ ²		
Self Serve Data Access		✓		✓	✓	
Data Masking				✓		
View reports in Cognos	✓ ³	✓ ³	✓ ³	✓ ³	✓ ³	✓ ³
IBM Hadoop Solution for Big Data		✓ ⁴		✓ ⁴	✓	✓

¹ Limited to 250 assets (any combination of glossary terms, categories, information governance policies and information governance rules)

² One database Source or Capture Agent excluding z/OS and must be used with DataStage as target

³ View only access for any pre-defined report provided for Information Server

⁴ Maximum of 5-node cluster Big Insights Standard Edition install in support of Information Server

⁵ Categorize up to 10 gigabytes of content

Separate add-on purchases: data replication, ERP connectors (SAP, SAS), Postal address verification / geo-coding